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HRS-7535 for Type 2 Diabetes Inadequately Controlled With Metformin
A Randomized Clinical Trial

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Abstract

IMPORTANCE Oral small-molecule glucagon-like peptide-1 receptor agonists (GLP-1 RAs) may address limitations of injectable and peptide-based agents for type 2 diabetes.

OBJECTIVE To evaluate the efficacy and safety of HRS-7535, an oral nonpeptide GLP-1 RA, as add-on therapy among adults with type 2 diabetes inadequately controlled with metformin.

DESIGN, SETTING, AND PARTICIPANTS This 16-week, phase 2, double-blind, placebo-controlled randomized clinical trial was conducted at 44 centers in China. Adults aged 18 to 75 years with type 2 diabetes, hemoglobin A_{1c} (HbA_{1c}) levels ranging from 7.5% to 11.0%, and receiving stable metformin therapy were enrolled between May 10 and December 18, 2023. Data were analyzed from May 8 to 22, 2024.

INTERVENTIONS Participants were randomized (1:1:1:1:1) to once-daily oral HRS-7535 (15, 30, 60, or 90 mg) or matching placebo. The 60- and 90-mg treatment groups used dose-escalation regimens.

MAIN OUTCOMES AND MEASURES The primary outcome was change in HbA_{1c} level from baseline to week 16. Secondary outcomes included changes in fasting plasma glucose level, 2-hour postprandial glucose level, body weight, and the proportion of patients achieving HbA_{1c} levels less than 7.0%. Safety outcomes included adverse events (AEs), hypoglycemia, and AEs of special interest.

RESULTS Of 194 randomized patients (mean [SD] age, 52.3 [11.0] years; 115 [59.3%] men; mean [SD] HbA_{1c} level, 8.5% [0.7]), 177 (91.2%) completed treatment. The mean changes in HbA_{1c} level at week 16 were −1.19% (95% CI, −1.54% to −0.84%) with the 15-mg dose, −1.59% (95% CI, −1.94% to −1.24%) with the 30-mg dose, −1.82% (95% CI, −2.16% to −1.48%) with the 60-mg dose, and −1.64% (95% CI, −1.97% to −1.30%) with the 90-mg dose compared with −0.25% (95% CI, −0.60% to 0.09%) with placebo; placebo-adjusted differences ranged from −0.94% (95% CI, −1.43% to −0.45%) to −1.57% (95% CI, −2.05% to −1.08%) (all $P < .001$). HbA_{1c} level less than 7.0% was achieved in proportions ranging from 48.7% (95% CI, 32.4%-65.2%) to 63.2% (95% CI, 46.0%-78.2%) of HRS-7535–treated patients vs 15.4% (95% CI, 5.9%-30.5%) with placebo. The 90-mg group had greater body weight reduction than placebo (−2.63% [95% CI, −3.72% to −1.54%] vs −1.30% [95% CI, −2.46% to −0.15%]). AEs occurred in 28 of 39 (71.8%) to 33 of 39 (84.6%) HRS-7535–treated patients and 28 of 39 (71.8%) placebo-treated patients and were predominantly mild to moderate gastrointestinal events. Level 1 hypoglycemia occurred in 9 HRS-7535–treated patients; no level 2 or 3 hypoglycemia, pancreatitis, or elevations of alanine aminotransferase or aspartate aminotransferase levels greater than 3 times the upper limit of normal occurred.

(continued)

Key Points

Question In adults with type 2 diabetes inadequately controlled with metformin, does once-daily oral HRS-7535 improve glycemic control compared with placebo?

Findings In this randomized clinical trial of 194 adults in China with type 2 diabetes receiving stable metformin, all HRS-7535 doses (15-90 mg once daily) reduced hemoglobin A_{1c} levels compared with placebo during a period of 16 weeks; placebo-adjusted differences ranged from −0.94% to −1.57%.

Meaning In this study, oral HRS-7535 added to metformin improved glycemic control and was generally well tolerated, supporting further phase 3 development.

+ Invited Commentary

+ Supplemental content

Author affiliations and article information are listed at the end of this article.

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Abstract (continued)

CONCLUSIONS AND RELEVANCE In this randomized clinical trial of adults with type 2 diabetes inadequately controlled with metformin, oral HRS-7535 improved glycemic control and was associated with modest weight reduction, with a safety profile consistent with that of GLP-1 RAs. Because HRS-7535 is a nonpeptide oral GLP-1 RA that does not require fasting administration or injection, it may be a viable treatment option, pending confirmation in phase 3 trials.

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Introduction

Glucagon-like peptide-1 receptor agonists (GLP-1 RAs) are established therapies for type 2 diabetes, providing effective glycemic control, weight reduction, and a low intrinsic risk of hypoglycemia.[1-3] Large cardiovascular outcome trials and meta-analyses[4-10] have also shown that GLP-1 RAs reduce major adverse cardiovascular events and slow progression of chronic kidney disease.

Oral GLP-1 RA formulations may improve treatment acceptance and adherence compared with injectable agents. Small-molecule oral agents may offer additional advantages, including simplified manufacturing and storage and the absence of cold-chain requirements.[11-15] Currently, the only approved oral GLP-1 RA is semaglutide, which was approved by the US Food and Drug Administration for type 2 diabetes in 2019.[16] However, the 14-mg oral dose is less effective than the 1-mg injectable formulation and requires a gastric permeation enhancer and strict fasting administration to optimize absorption.[17] These limitations support development of orally available, once-daily, nonpeptide GLP-1 RAs without such constraints.

Several oral small-molecule GLP-1 RAs have entered clinical development, including orforglipron[18] and lotiglipron.[19] In phase 3 trials, orforglipron reduced hemoglobin A_{1c} (HbA_{1c}) levels compared with placebo in adults with early type 2 diabetes and was noninferior and superior to oral semaglutide for reduction of HbA_{1c} levels in adults with type 2 diabetes inadequately controlled with metformin.[20,21] In contrast, lotiglipron development was discontinued because of elevations of liver transaminase levels.[22]

HRS-7535 is a novel oral small-molecule GLP-1 RA that has shown high bioavailability and favorable pharmacodynamic effects in preclinical studies, including improved glucose tolerance, enhanced insulin secretion, reduced appetite, and reduced food intake. In a first-in-human phase 1 trial,[23] (NCT05347758), HRS-7535 showed a safety and tolerability profile consistent with that of other GLP-1 RAs and pharmacokinetic characteristics supportive of once-daily dosing. Multiple-dose titration regimens (30, 60, 90, and 120 mg) also produced clinically relevant lowering of glucose levels and substantial short-term weight loss in healthy participants.[23] In this phase 2 trial, we evaluated the efficacy and safety of once-daily oral HRS-7535 at 4 dose levels compared with placebo as add-on therapy to metformin among adults in China with type 2 diabetes inadequately controlled with metformin monotherapy during a period of 16 weeks.

Methods

Study Design

This phase 2, double-blind, placebo-controlled, multicenter randomized clinical trial was conducted at 44 sites in China and included a 2-week screening period, a 3-week run-in period, a 16-week double-blind treatment period, and safety follow-up. The study was conducted in accordance with the Declaration of Helsinki,[24] International Council for Harmonisation guidelines, and applicable local regulations. The protocol and amendments (Supplement 1) were approved by the institutional

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review board or independent ethics committee at each site, and all participants provided written informed consent. The trial followed the Consolidated Standards of Reporting Trials (CONSORT) reporting guideline.

Participants

Eligible participants ranged in age from 18 to 75 years, weighed at least 50 kg, had a body mass index (BMI; calculated as weight in kilograms divided by height in meters squared) of 19.0 to 40.0, and had type 2 diabetes with an HbA_{1c} level of 7.5% to 11.0% (to convert to a proportion of hemoglobin, multiply by 0.01), despite stable metformin therapy (≥1000 mg/d for ≥8 weeks) and stable body weight (±5 kg) for at least 3 months before screening. Additional eligibility criteria are provided in the trial protocol (Supplement 1).

Procedures

Eligible participants were enrolled between May 10 and December 18, 2023, through a centralized interactive response technology system and randomly assigned in a 1:1:1:1:1 ratio to once-daily oral HRS-7535 (15, 30, 60, or 90 mg) or matching placebo through a centralized interactive web response system. Treatment allocation was implemented through unique kit numbers and concealed from investigators and site staff throughout the trial. The 60- and 90-mg groups followed prespecified titration regimens (eFigure 1 in Supplement 2), both starting at the 30-mg dose; the 60-mg group increased the dosage by 15 mg weekly for 2 weeks and then continued the 60-mg dosage, and the 90-mg group increased by 30 mg every 4 weeks until reaching 90 mg. Participants, investigators, study staff, and data analysts remained blinded throughout the trial.

End Points

The primary end point was the change from baseline in HbA_{1c} level at week 16, measured by a central laboratory. Secondary end points included safety and tolerability; changes from baseline at week 16 in levels of fasting plasma glucose (FPG), fasting serum insulin, fasting C-peptide, 2-hour postprandial plasma glucose (2h-PPG), 2-hour postprandial serum insulin, and 2-hour postprandial serum C-peptide, body weight, waist circumference, and mean 7-point self-monitored blood glucose (SMBG) level; the proportion of patients achieving an HbA_{1c} level of less than 7.0%; the proportion requiring rescue antihyperglycemic therapy; and the pharmacokinetic profile of once-daily oral HRS-7535 at week 16.

Safety assessments included adverse events (AEs), hypoglycemic events, vital signs, physical examinations, electrocardiography, and laboratory parameters. Participants were instructed to record capillary glucose values of 3.9 mmol/L (70 mg/dL) or lower and any symptoms suggestive of hypoglycemia; definitions are provided in eTable 1 in Supplement 2. Prespecified AEs of special interest included liver function abnormalities, elevated pancreatic enzyme levels, pancreatitis, elevated calcitonin levels, and severe hypoglycemia.

Statistical Analysis

Data were analyzed from May 8 to 22, 2024. Efficacy analyses were conducted in the full analysis set, which comprised all randomized patients who received at least 1 dose of study treatment. Safety analyses included all patients who received at least 1 dose of study treatment.

A sample size of 30 patients per group provided more than 90% power to detect a difference of 0.85 percentage points in reduction of HbA_{1c} level at week 16 between any HRS-7535 group and the placebo group, assuming an SD of 1.0% and a 2-sided type I error rate of .05. With 20% dropout, 36 patients were planned per group (180 total). No formal adjustment for multiplicity was applied.

The primary estimand used a hypothetical strategy under the International Council for Harmonisation E9(R1) framework.[25] Intercurrent events were premature discontinuation of study treatment or initiation of rescue antihyperglycemic medication before week 16; data collected thereafter were excluded, and missing data were assumed to be missing at random. Change from

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baseline in HbA_{1c} level at week 16 was analyzed using a mixed-effects model for repeated measures (MMRM) with baseline HbA_{1c} level as a covariate and treatment group, visit, and treatment-by-visit interaction as fixed effects, patient as a random effect, and an unstructured covariance matrix. Missing data not attributable to intercurrent events were handled implicitly within the MMRM under a missing-at-random assumption. A treatment-policy sensitivity analysis used copy-reference multiple imputation, and an ad hoc analysis excluded patients without measurable postbaseline pharmacokinetic exposure.

Secondary efficacy end points were analyzed under the same hypothetical estimand. Changes from baseline in FPG level, body weight, percentage body weight, waist circumference, and 7-point SMBG level at week 16 were analyzed using the same MMRM approach, with the corresponding baseline value as a covariate. Changes in levels of fasting serum insulin, fasting C-peptide, 2h-PPG, 2-hour postprandial serum insulin, and 2-hour postprandial serum C-peptide were analyzed using analysis of covariance with the corresponding baseline value as a covariate; missing data were imputed using the last observation carried forward. For HbA_{1c} levels less than 7.0% at week 16, missing week 16 HbA_{1c} values were imputed under a missing-at-random assumption and binary response status was derived from the imputed continuous values.

All analyses were performed using SAS, version 9.4 (SAS Institute Inc). Additional details are provided in the statistical analysis plan (Supplement 1). Two-sided $P < .05$ indicated statistical analysis.

Results

Patients

A total of 194 patients were randomized and received at least 1 dose of study treatment. Of these, 177 patients (91.2%; range across groups, 33 of 39 [84.6%] for the 30-mg group to 38 of 39 [97.4%] for the placebo group) completed the 16-week treatment period (**Figure 1**). The most common reasons for discontinuation were patient decision (8 [4.1%]) and AEs (6 [3.1%]). Baseline demographic and clinical characteristics were generally well balanced across treatment groups (**Table 1**). The mean (SD) age was 52.3 (11.0) years, the mean (SD) baseline HbA_{1c} level was 8.5% (0.7%), and the mean (SD) BMI was 26.7 (3.8). Seventy-nine patients (40.7%) were female and 115 (59.3%) were male.

Efficacy

At week 16, least-squares (LS) mean changes from baseline in HbA_{1c} levels were −1.19% (95% CI, −1.54% to −0.84%) in the 15-mg group, −1.59% (95% CI, −1.94% to −1.24%) in the 30-mg group, −1.82% (95% CI, −2.16% to −1.48%) in the 60-mg group, and −1.64% (95% CI, −1.97% to −1.30%) in the 90-mg group compared with −0.25% (95% CI, −0.60% to 0.09%) in the placebo group (**Table 2** and **Figure 2**A). All HRS-7535 doses were superior to placebo, with placebo-adjusted LS mean differences of −0.94% (95% CI, −1.43% to −0.45%) in the 15-mg group, −1.34% (95% CI, −1.83% to −0.84%) in the 30-mg group, −1.57% (95% CI, −2.05% to −1.08%) in the 60-mg group, and −1.39% (95% CI, −1.87% to −0.91%) in the 90-mg group (all $P < .001$). Reductions in HbA_{1c} levels with HRS-7535 were consistently greater than those with placebo across most prespecified subgroups (eTable 2 in Supplement 2). These results were consistent in a treatment-policy sensitivity analysis using copy-reference multiple imputation and in an ad hoc analysis excluding patients without measurable postbaseline pharmacokinetic exposure (eTables 3 and 4 in Supplement 2).

A greater proportion of patients receiving HRS-7535 achieved HbA_{1c} levels less than 7.0% at week 16 (range, 48.7% [95% CI, 32.4%-65.2%] to 63.2% [95% CI, 46.0%-78.2%]) than those receiving placebo (15.4% [95% CI, 5.9%-30.5%]) (all $P \leq .002$) (Table 2 and Figure 2B). Reductions in levels of FPG, 2-hour PPG, and mean 7-point SMBG were also greater with all HRS-7535 doses than with placebo (Table 2, Figure 2C, and eFigure 2 in Supplement 2).

For body weight, the least-squares mean change from baseline at week 16 was −2.04 kg (95% CI, −2.84 to −1.24 kg) in the 90-mg group vs −1.06 kg (95% CI, −1.91 to −0.22 kg) with placebo,

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corresponding to percentage changes of −2.63% and −1.30%, respectively (Table 2, Figure 2D, and eFigure 3 in Supplement 2). Changes in body weight with lower HRS-7535 doses were modest and generally similar to those with placebo. Changes in waist circumference were small across groups, although numerically greater reductions were observed with higher HRS-7535 doses than with placebo (Table 2 and eFigure 4 in Supplement 2).

At week 16, fasting serum insulin and fasting C-peptide levels were higher with HRS-7535 90 mg than with placebo. HRS-7535 doses of 30 mg or higher were associated with greater reductions in 2-hour postprandial serum insulin level. Fasting C-peptide level also increased in the 30-mg group. No clinically meaningful differences in 2-hour postprandial serum C-peptide level were observed across HRS-7535 dose groups vs placebo (Table 2). Rescue antihyperglycemic therapy for hyperglycemia was required in 1 of 39 patients (2.6%) in the 15-mg group, 3 of 39 (7.7%) in the 30-mg group, 0 of 38 in the 60-mg group, and 1 of 39 (2.6%) in the 90-mg group compared with 12 of 39 (30.8%) in the placebo group.

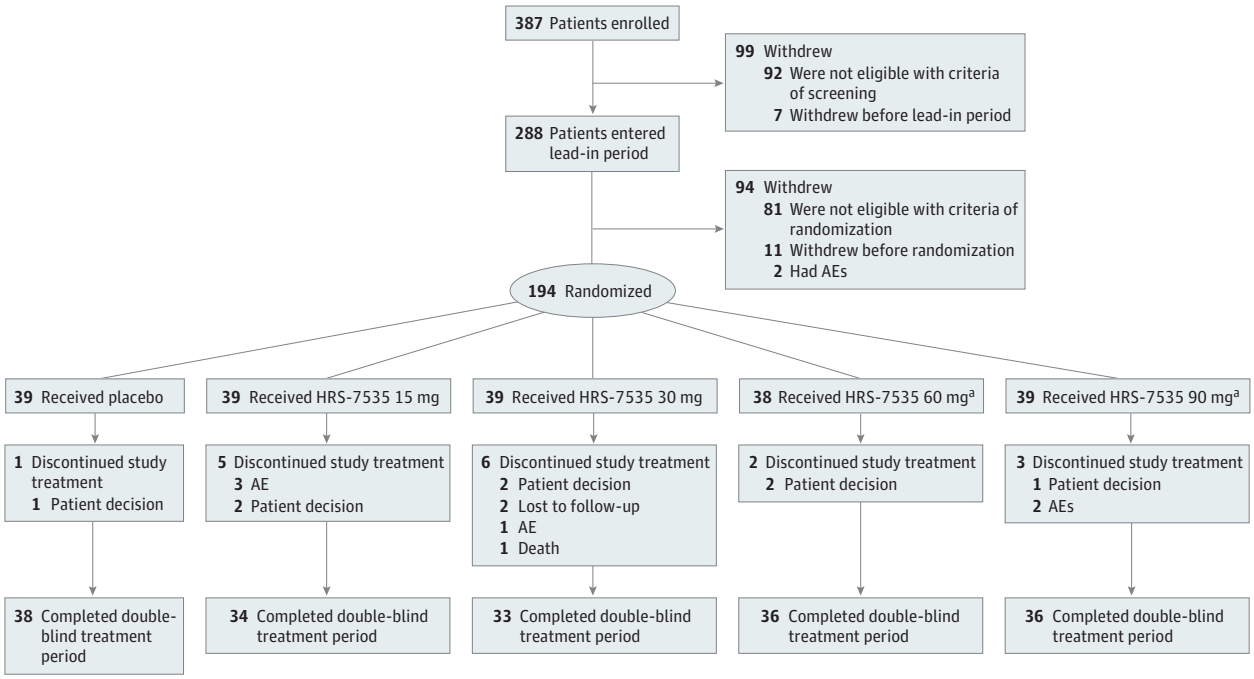
Pharmacokinetics

Predose plasma trough concentrations of HRS-7535 are shown in eTable 5 in Supplement 2 and generally increased with dose. At week 16, mean (SD) trough concentrations were 42.1 (85.9) ng/mL in the 15-mg group, 59.0 (53.2) ng/mL in the 30-mg group, 128 (128) ng/mL in the 60-mg group, and 297 (359) ng/mL in the 90-mg group.

Safety

Overall, AEs were reported in 28 of 39 patients (71.8%) in the placebo group, 31 of 39 (79.5%) in the 15-mg group, 28 of 39 (71.8%) in the 30-mg group, 29 of 38 (76.3%) in the 60-mg group, and 33 of 39 (84.6%) in the 90-mg group (**Table 3**). Gastrointestinal AEs were the most common treatment-related events and occurred in 2 of 39 (5.1%) patients in the placebo group and in 15 of 39 (38.5%) in

Figure 1. CONSORT Diagram



AE indicates adverse events.

[a] Participants in the HRS-7535 60-mg and 90-mg groups followed prespecified titration regimens. Both groups started at a dose of 30 mg once daily. In the 60-mg group, the dose was increased by 15 mg weekly for 2 weeks and then maintained at 60 mg. In the 90-mg group, the dose was increased by 30 mg every 4 weeks until 90 mg was reached and then maintained for the remainder of the 16-week treatment period.

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the 15-mg group, 9 of 39 (23.1%) in the 30-mg group, 17 of 38 (44.7%) in the 60-mg group, and 20 of 39 (51.3%) in the 90-mg group (eTable 6 in Supplement 2). Among the 63 patients with gastrointestinal AEs, 62 had events of mild or moderate severity; 1 patient in the 60-mg group reported severe diarrhea during titration at 30 mg. Nausea occurred in 1 of 39 patients (2.6%) in the placebo group, 3 of 39 (7.7%) in the 15-mg group, 3 of 39 (7.7%) in the 30-mg group, 13 of 38 (34.2%) in the 60-mg group, and 8 of 39 (20.5%) in the 90-mg group. Vomiting occurred in 0 patients in the placebo group, 1 of 39 (2.6%) in the 15-mg group, 2 of 39 (5.1%) in the 30-mg group, 6 of 38 (15.8%) in the 60-mg group, and 5 of 39 (12.8%) in the 90-mg group. Diarrhea occurred in 0 patients in the placebo group, 3 of 39 (7.7%) in the 15-mg group, 2 of 39 (5.1%) in the 30-mg group, 3 of 38 (7.9%) in the 60-mg group, and 6 of 39 (15.4%) in the 90-mg group (Table 3). Most gastrointestinal AEs occurred early during treatment or at dose escalation. Daily gastrointestinal AE rates were higher in the 60-mg group than in the 90-mg group during days 14 to 28 and days 28 to 56, suggesting that titration every 2 weeks may not have allowed sufficient time for gastrointestinal tolerability to develop (eFigure 5 in Supplement 2).

Table 1. Baseline Demographic and Clinical Characteristics

Characteristic	Treatment group				
		HRS-7535 dose once daily			
	Placebo (n = 39)	15 mg (n = 39)	30 mg (n = 39)	60 mg (n = 38)	90 mg (n = 39)
Age, mean (SD), y	48.9 (10.4)	52.4 (11.1)	55.5 (9.4)	52.2 (12.2)	52.4 (11.4)
Sex, No. (%)					
Female	16 (41.0)	17 (43.6)	13 (33.3)	14 (36.8)	19 (48.7)
Male	23 (59.0)	22 (56.4)	26 (66.7)	24 (63.2)	20 (51.3)
Body weight, mean (SD), kg	75.3 (13.3)	72.6 (13.3)	69.1 (9.3)	75.3 (12.7)	74.9 (12.6)
BMI, mean (SD)	27.1 (4.0)	26.3 (4.0)	25.2 (2.8)	27.1 (3.5)	27.9 (4.0)
BMI, No. (%)					
≥26	25 (64.1)	14 (35.9)	15 (38.5)	22 (57.9)	24 (61.5)
<26	14 (35.9)	25 (64.1)	24 (61.5)	16 (42.1)	15 (38.5)
Waist circumference, mean (SD), cm	95.4 (10.2)	94.0 (7.7)	93.3 (8.6)	95.9 (8.8)	96.5 (8.9)
Duration of type 2 diabetes, mean (SD), y	5.2 (5.1)	5.3 (4.0)	5.1 (4.1)	3.7 (3.9)	5.5 (5.1)
Background metformin extended-release dose, No. (%)[a]					
1000 mg/d	10 (25.6)	8 (20.5)	8 (20.5)	4 (10.5)	10 (25.6)
1500 mg/d	18 (46.2)	22 (56.4)	21 (53.8)	24 (63.2)	22 (56.4)
2000 mg/d	9 (23.1)	9 (23.1)	9 (23.1)	9 (23.7)	5 (12.8)
Other	2 (5.1)	0	1 (2.6)	1 (2.6)	2 (5.1)
HbA_{1c} level, mean (SD), %	8.4 (0.6)	8.5 (0.7)	8.6 (0.8)	8.5 (0.7)	8.6 (0.8)
HbA_{1c} level, No. (%)					
>8.5%	15 (38.5)	15 (38.5)	18 (46.2)	12 (31.6)	18 (46.2)
≤8.5%	24 (61.5)	24 (61.5)	21 (53.8)	26 (68.4)	21 (53.8)
FPG, mean (SD), mg/dL[b]	182.0 (30.6)	174.8 (36.0)	178.4 (36.0)	174.8 (41.4)	185.6 (43.2)
Fasting serum insulin, mean (SD), μIU/mL[b]	13.6 (8.2)	12.0 (6.1)	10.4 (5.9)	14.3 (12.0)	14.0 (8.1)
Fasting C-peptide, mean (SD), ng/mL[b]	2.8 (1.0)	2.7 (0.9)	2.4 (0.9)	2.7 (0.8)	2.8 (1.0)
2-h PPG, mean (SD), mg/dL[b]	291.9 (59.5)	288.3 (61.3)	304.5 (55.9)	288.3 (59.5)	286.5 (64.9)
SBP, mean (SD), mm Hg	123.3 (12.4)	122.9 (11.8)	125.0 (11.8)	122.8 (9.4)	123.6 (10.2)
DBP, mean (SD), mm Hg	81.3 (9.2)	80.8 (8.1)	81.1 (8.7)	79.7 (9.3)	80.5 (6.8)
eGFR, mean (SD), mL/min/1.73 m^2	107.2 (14.0)	107.3 (14.6)	101.5 (14.9)	104.3 (12.9)	105.0 (15.2)

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; FPG, fasting plasma glucose; HbA1c, hemoglobin A1c; PPG, postprandial plasma glucose; SBP, systolic blood pressure.

SI conversion factors: To convert C-peptide to nmol/L, multiply by 0.331; FPG to mmol/L, multiply by 0.0555; HbA_{1c} to a proportion of hemoglobin, multiply by 0.01; insulin to pmol/L, multiply by 6.945; and PPG to mmol/L, multiply by 0.0555.

[a] To maintain consistency of background therapy, metformin was standardized to extended-release metformin (Glucophage XR) during the run-in and double-blind periods at 1500 mg/d or 2000 mg/d according to prespecified criteria: if the original dose was greater than 1000 mg/d and less than 1700 mg/d, the dose was adjusted to 1500 mg/d; if 1700 mg/d or greater, to 2000 mg/d; and if 1000 mg/d, the dose remained unchanged.

[b] Local laboratory testing was used for measurements.

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Table 2. Primary and Secondary Efficacy End Points at Week 16

End point	Treatment group				
		HRS-7535 dose once daily			
	Placebo (n = 39)	15 mg (n = 39)	30 mg (n = 39)	60 mg (n = 38)	90 mg (n = 39)
HbA1c level, %					
No. with observed value at week 16	26	33	30	36	35
LS change from baseline, mean (95% CI)	−0.25 (−0.60 to 0.09)	−1.19 (−1.54 to −0.84)	−1.59 (−1.94 to −1.24)	−1.82 (−2.16 to −1.48)	−1.64 (−1.97 to −1.30)
LS mean difference vs placebo (95% CI)	NA	−0.94 (−1.43 to −0.45)	−1.34 (−1.83 to −0.84)	−1.57 (−2.05 to −1.08)	−1.39 (−1.87 to −0.91)
P value[a]	NA	<.001	<.001	<.001	<.001
FPG level, mg/dL					
No. with observed value at week 16	26	33	30	36	35
LS change from baseline, mean (95% CI)	−2.52 (−14.05 to 9.01)	−24.68 (−35.85 to −13.69)	−27.75 (−39.10 to −16.40)	−48.47 (−59.10 to −37.84)	−34.59 (−45.22 to −23.96)
LS mean difference vs placebo (95% CI)	NA	−22.16 (−38.20 to −6.13)	−25.23 (−41.44 to −9.01)	−45.95 (−61.62 to −30.27)	−32.07 (−47.75 to −16.40)
P value[a]	NA	.007	.003	<.001	<.001
Fasting serum insulin level, μIU/mL					
No. with observed value at week 16	39	39	39	38	39
LS change from baseline, mean (95% CI)	−0.55 (−2.87 to 1.78)	0.41 (−1.91 to 2.74)	1.90 (−0.44 to 4.24)	0.04 (−2.32 to 2.40)	2.96 (0.64 to 5.29)
LS mean difference vs placebo (95% CI)	NA	0.96 (−2.33 to 4.25)	2.45 (−0.86 to 5.75)	0.58 (−2.72 to 3.89)	3.51 (0.23 to 6.79)
P value[b]	NA	.57	.15	.73	.04
Fasting C-peptide level, ng/mL					
No. with observed value at week 16	39	39	39	38	39
LS change from baseline, mean (95% CI)	−0.11 (−0.37 to 0.15)	0.06 (−0.20 to 0.32)	0.35 (0.09 to 0.61)	0.09 (−0.17 to 0.36)	0.50 (0.24 to 0.76)
LS mean difference vs placebo (95% CI)	NA	0.17 (−0.20 to 0.54)	0.46 (0.09 to 0.83)	0.21 (−0.17 to 0.58)	0.61 (0.24 to 0.98)
P value[b]	NA	.36	.02	.28	.001
2-hour PPG level, mg/dL					
No. with observed value at week 16	39	39	39	38	39
LS change from baseline, mean (95% CI)	− 1.62 (−26.31 to 23.24)	−99.64 (−124.50 to −74.95)	−114.23 (−139.10 to −89.37)	−152.25 (−177.48 to −127.21)	−131.35 (−156.04 to −106.49)
LS mean difference vs placebo (95% CI)	NA	−98.02 (−133.15 to −63.06)	−112.61 (−147.75 to −77.48)	−150.81 (−185.95 to −115.50)	−129.73 (−164.68 to −94.77)
P value[b]	NA	<.001	<.001	<.001	<.001
2-h Postprandial serum insulin level, μIU/mL					
No. with observed value at week 16	39	39	39	38	39
LS change from baseline, mean (95% CI)	3.5 (−2.3 to 9.4)	0.7 (−5.1 to 6.6)	−6.5 (−12.3 to −0.6)	−7.5 (−13.4 to −1.6)	−12.1 (−17.9 to −6.3)
LS mean difference vs placebo (95% CI)	NA	−2.8 (−11.1 to 5.5)	−10.0 (−18.3 to −1.7)	−11.0 (−19.3 to −2.7)	−15.6 (−23.9 to −7.4)
P value[b]	NA	.51	.02	.01	<.001
2-h Postprandial serum C-peptide level, ng/mL					
No. with observed value at week 16	39	39	39	38	39
LS change from baseline, mean (95% CI)	0.37 (−0.24 to 0.98)	0.68 (0.07 to 1.29)	0.21 (−0.40 to 0.82)	−0.39 (−1.01 to 0.23)	−0.25 (−0.86 to 0.36)
LS mean difference vs placebo (95% CI)	NA	0.31 (−0.55 to 1.18)	−0.16 (−1.02 to 0.71)	−0.76 (−1.63 to 0.11)	−0.62 (−1.48 to 0.25)
P value[b]	NA	.47	.72	.08	.16
Fasting body weight, kg					
No. with observed value at week 16	26	33	30	36	35
LS change from baseline, mean (95% CI)	−1.06 (−1.91 to −0.22)	−0.33 (−1.16 to 0.51)	−0.42 (−1.27 to 0.42)	−1.03 (−1.84 to −0.23)	−2.04 (−2.84 to −1.24)
LS mean difference vs placebo (95% CI)	NA	0.74 (−0.45 to 1.92)	0.64 (−0.56 to 1.84)	0.03 (−1.14 to 1.20)	−0.97 (−2.13 to 0.19)
P value[b]	NA	.22	.29	.96	.10
Change in fasting body weight, %					
No. with observed value at week 16	33	30	36	35	26
LS change from baseline, mean (95% CI)	−1.30 (−2.46 to −0.15)	−0.40 (−1.53 to 0.73)	−0.50 (−1.65 to 0.66)	−1.49 (−2.59 to −0.40)	−2.63 (−3.72 to −1.54)
LS mean difference vs placebo (95% CI)	NA	0.90 (−0.71 to 2.52)	0.81 (−0.83 to 2.44)	−0.19 (−1.78 to 1.40)	−1.32 (−2.91 to 0.26)
P value[b]	NA	.27	.33	.82	.10

(continued)

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Table 2. Primary and Secondary Efficacy End Points at Week 16 (continued)

End point	Treatment group				
		HRS-7535 dose once daily			
	Placebo (n = 39)	15 mg (n = 39)	30 mg (n = 39)	60 mg (n = 38)	90 mg (n = 39)
Waist circumference, cm					
No. with observed value at week 16	26	33	30	36	35
LS change from baseline, mean (95% CI)	−1.31 (−2.38 to −0.24)	−0.71 (−1.75 to 0.33)	−1.25 (−2.31 to −0.19)	−1.95 (−2.95 to −0.94)	−2.70 (−3.70 to −1.70)
LS mean difference vs placebo (95% CI)	NA	0.60 (−0.89 to 2.09)	0.06 (−1.45 to 1.57)	−0.64 (−2.10 to 0.83)	−1.38 (−2.85 to 0.08)
P value[b]	NA	.43	.94	.39	.06
7-Point SMBG level, mg/dL					
No. with observed value at week 16	26	31	28	34	33
LS change from baseline, mean (95% CI)	−15.86 (−28.47 to −3.24)	−34.95 (−47.03 to −22.88)	−46.67 (−59.28 to −34.0)	−58.56 (−70.27 to −47.03)	−52.25 (−63.96 to −40.54)
LS mean difference vs placebo (95% CI)	NA	−19.10 (−36.58 to −1.62)	−30.81 (−48.65 to −12.79)	−42.70 (−59.82 to −25.59)	−36.22 (−53.51 to −19.10)
P value[b]	NA	.03	<.001	<.001	<.001
Patients with HbA_{1c} level <7.0%					
No./Total No. of patients[c]	6/39	19/39	22/39	24/38	20/39
Proportion of patients (95% CI), %[d]	15.4 (5.9 to 30.5)	48.7 (32.4 to 65.2)	56.4 (39.6 to 72.2)	63.2 (46.0 to 78.2)	51.3 (34.8 to 67.6)
Difference vs placebo (95% CI)[e]	NA	33.3 (14.0 to 52.7)	41.0 (21.8 to 60.3)	47.8 (28.7 to 66.8)	35.9 (16.6 to 55.2)
P value[f]	NA	.002	<.001	<.001	<.001

Abbreviations: FPG, fasting plasma glucose; HbA_{1c}, hemoglobin A_{1c}; LS, least squares; PPG, postprandial plasma glucose; NA, not applicable; SMBG, self-monitored blood glucose.

SI conversion factors: To convert C-peptide to nmol/L, multiply by 0.331; FPG to mmol/L, multiply by 0.0555; HbA_{1c} to a proportion of hemoglobin, multiply by 0.01; insulin to pmol/L, multiply by 6.945; PPG to mmol/L, multiply by 0.0555; and SMBG to mmol/L, multiply by 0.0555.

[a] HRS-7535 dose groups were compared with placebo using a mixed-effects model for repeated measures based on restricted maximum likelihood. The model included baseline values as covariates and treatment group, visit, and treatment-by-visit interaction as fixed effects, with patient as a random effect and an unstructured covariance matrix.

[b] HRS-7535 dose groups were compared with placebo using analysis of covariance models, with treatment group as the independent variable and baseline values as covariates; missing data, including those affected by intercurrent events, were imputed with last observation carried forward.

[c] Indicates number of patients achieving an HbA_{1c} level less than 7.0% at week 16 after multiple imputation under the missing-at-random assumption and the total number of randomized patients.

[d] 95% CIs for proportions were calculated using the Clopper-Pearson method.

[e] 95% CIs for between-group differences in proportions were calculated using normal approximation.

[f] The proportions of patients achieving an HbA_{1c} level less than 7.0% at week 16 were compared between each HRS-7535 dose group and placebo using χ^2 tests.

Documented hypoglycemic events occurred in 1 of 39 patients (2.6%) in the 15-mg group, 2 of 39 (5.1%) in the 30-mg group, 4 of 38 (10.5%) in the 60-mg group, and 2 of 39 (5.1%) in the 90-mg group and in none of the placebo-treated patients. All events were level 1 hypoglycemia; no level 2 or level 3 events were reported (eTable 7 in Supplement 2).

Prespecified AEs of special interest occurred in 5 patients receiving HRS-7535 and in no patients receiving placebo; all were lipase level elevations greater than 3 times the upper limit of normal (Table 3). No liver function–related AEs of special interest, pancreatitis, abnormal blood calcitonin, or severe hypoglycemia were observed.

Treatment discontinuation due to AEs occurred in 3 of 39 patients (7.7%) in the 15-mg group, 2 of 39 (5.1%) in the 30-mg group, 0 of 38 in the 60-mg group, and 2 of 39 (5.1%) in the 90-mg group and in none of the placebo-treated patients (eTable 8 in Supplement 2). Serious AEs were reported in 0 to 2 patients per HRS-7535 group (0%-5.1%) and in 2 of 39 (5.1%) in the placebo-treated group (eTable 9 in Supplement 2). One patient in the 30-mg group died of cerebral infarction; the investigator assessed the event as unrelated to study treatment.

Dose-related increases in amylase and lipase levels were observed with HRS-7535 compared with placebo (eTable 10 in Supplement 2). Small mean (SD) increases in calcitonin levels were also observed across HRS-7535 groups (range, 12.2% [33.6%] to 31.3% [93.9%]) compared with placebo (9.9% [35.5%]), with no cases meeting the prespecified criteria for calcitonin-related AEs of special

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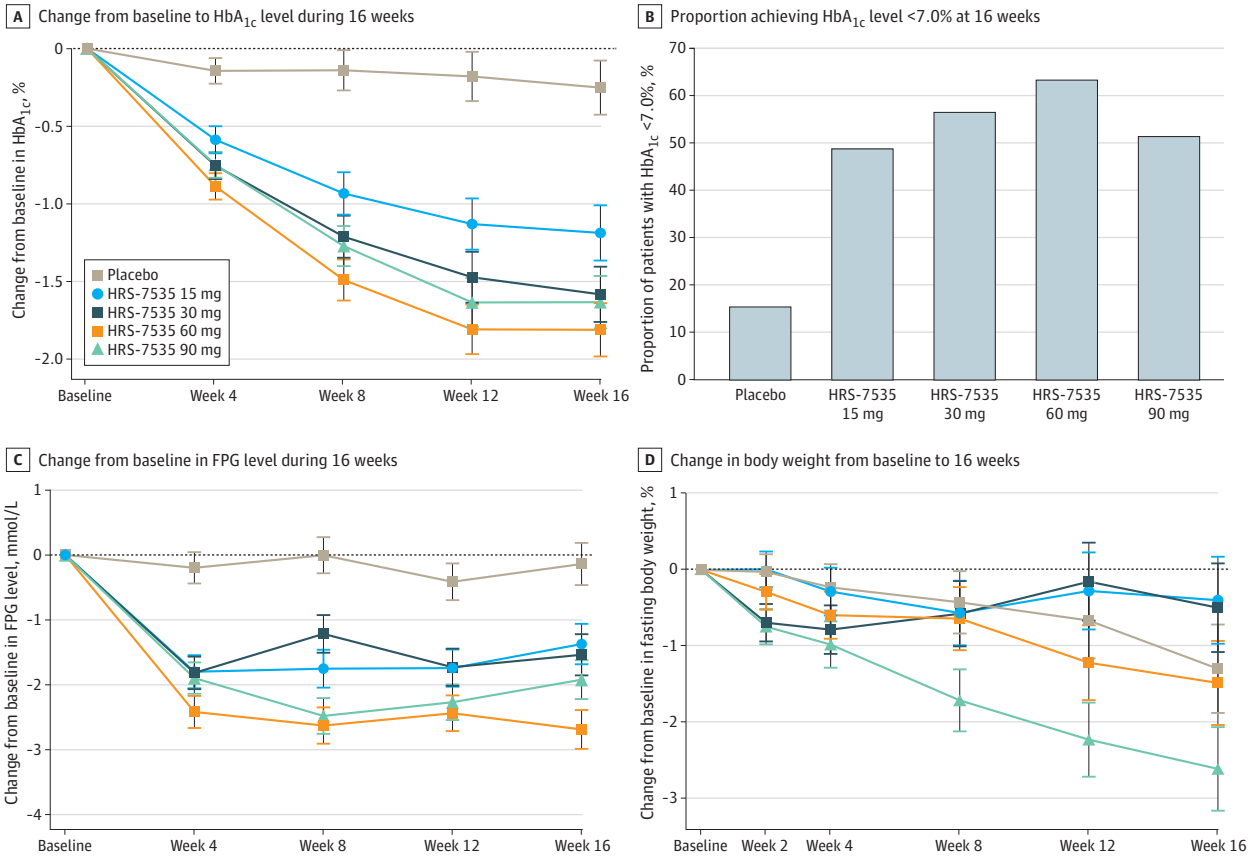
interest. No patients receiving HRS-7535 had alanine aminotransferase or aspartate aminotransferase elevations greater than 3 times the upper limit of normal.

At week 16, the mean (SD) pulse rate increased by 2.7 (9.9) to 3.2 (8.1) beats per minute (bpm) in the 30-mg and 90-mg groups compared with 0.8 (8.0) bpm in the placebo group, and mean (SD) heart rate increased by 3.0 (8.9) to 5.4 (9.4) bpm in the 30-mg, 60-mg, and 90-mg groups compared with 0.9 (9.1) bpm in the placebo group (eTable 11 in Supplement 2). No clinically relevant changes were observed in other laboratory measures, vital signs, physical examination findings, or electrocardiographic results (eTables 10-12 in Supplement 2).

Discussion

In this randomized clinical trial, once-daily oral HRS-7535 added to metformin reduced HbA_{1c} levels in Chinese adults with type 2 diabetes inadequately controlled with metformin monotherapy. All 4 HRS-7535 doses were superior to placebo, with placebo-adjusted reductions of HbA_{1c} levels ranging from −0.94% to −1.57%, and 48.7% to 63.2% of patients achieved an HbA_{1c} level less than 7.0% compared with 15.4% in the placebo group. Reduction of HbA_{1c} levels with HRS-7535 60 mg was similar to that with 90 mg, suggesting that the lower dose may provide robust glycemic control. The

Figure 2. Line and Bar Graphs of Main Efficacy End Points at Week 16



Data are presented as least-squares mean (SE) for the line graphs in panels A, C, and D and as percentages for the bar graph in panel B. HbA_{1c} indicates hemoglobin A_{1c}. Error bars indicate 95% CIs.

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overall pattern of HbA_{1c} level reduction was broadly consistent with that reported for other oral nonpeptide GLP-1 receptor agonists, including orforglipron.[20,21,26]

In addition to glycemic improvement, HRS-7535 90 mg was associated with modest weight reduction compared with placebo, with a continued downward trajectory and no clear plateau at week 16. Lower doses were associated with smaller changes in body weight. Weight reduction in this trial was less than that reported in the earlier phase 1 multiple-dose study,[22] in which body weight decreased by 4.38 kg (6.63%) after 28 days of more intensive titration (30, 60, 90, and 120 mg). Several factors may have contributed to this difference, including shorter exposure to the target dose in our phase 2 trial (maximum, 8 weeks at 90 mg), lower baseline body weight, and the lower BMI typical of Chinese populations compared with the largely White populations enrolled in other trials of oral small-molecule GLP-1 receptor agonists.[18,26,27] A phase 2 study[28] evaluated higher maintenance doses of HRS-7535 (≤180 mg daily) in Chinese adults with obesity inadequately controlled by lifestyle intervention; results may further inform the weight reduction profile of this agent.

HRS-7535 added to metformin for 16 weeks was generally safe and well tolerated, with no new safety signals identified. The overall AE profile was consistent with that of other GLP-1 RAs, with predominantly gastrointestinal events that were mostly mild or moderate in severity and occurred mainly during treatment initiation or dose escalation.[11,12,17,18,22,29-31] In our trial, titration every 2 weeks may not have allowed sufficient time for gastrointestinal toleration to develop. Treatment discontinuation rates were low and, despite the shorter treatment duration, appeared numerically lower than those reported in a 26-week phase 2 trial of orforglipron.[18] The incidence of hypoglycemia

Table 3. AEs During the 16-Week Treatment Period

Event	Treatment group, No. (%)				
		HRS-7535 dose once daily			
	Placebo (n = 39)	15 mg (n = 39)	30 mg (n = 39)	60 mg (n = 38)	90 mg (n = 39)
≥1 AE	28 (71.8)	31 (79.5)	28 (71.8)	29 (76.3)	33 (84.6)
AE related to study drug	7 (17.9)	23 (59.0)	16 (41.0)	21 (55.3)	23 (59.0)
Serious AE	2 (5.1)	0	1 (2.6)	1 (2.6)	1 (2.6)
Death[a]	0	0	1 (2.6)	0	0
Discontinuations following an AE	0	3 (7.7)	2 (5.1)	0	2 (5.1)
Prespecified AESI[b]	0	2 (5.1)	1 (2.6)	1 (2.6)	1 (2.6)
Liver function abnormality[c]	0	0	0	0	0
Amylase and/or lipase level >3 times ULN	0	2 (5.1)	1 (2.6)	1 (2.6)	1 (2.6)
Acute and/or chronic pancreatitis	0	0	0	0	0
Blood calcitonin level abnormal[d]	0	0	0	0	0
Episodes of severe hypoglycemia[e]	0	0	0	0	0
AEs occurring in ≥10% patients in any group					
Decreased appetite	1 (2.6)	8 (20.5)	4 (10.3)	5 (13.2)	9 (23.1)
Nausea	1 (2.6)	3 (7.7)	3 (7.7)	13 (34.2)	8 (20.5)
Upper respiratory tract infection	8 (20.5)	2 (5.1)	4 (10.3)	4 (10.5)	6 (15.4)
Diarrhea	0	3 (7.7)	2 (5.1)	3 (7.9)	6 (15.4)
Vomiting	0	1 (2.6)	2 (5.1)	6 (15.8)	5 (12.8)
Lipase level increased	0	3 (7.7)	5 (12.8)	5 (13.2)	5 (12.8)
Sinus tachycardia	1 (2.6)	4 (10.3)	1 (2.6)	3 (7.9)	4 (10.3)
Hyperuricemia	3 (7.7)	2 (5.1)	1 (2.6)	2 (5.3)	4 (10.3)
Hyperlipidemia	4 (10.3)	5 (12.8)	4 (10.3)	1 (2.6)	3 (7.7)
Urine albumin-creatinine ratio increased	3 (7.7)	5 (12.8)	3 (7.7)	1 (2.6)	2 (5.1)

Abbreviations: AE, adverse event; AESI, AE of special interest; ALT, alanine aminotransferase; TBIL, total bilirubin; ULN, upper limit of normal.

[a] The death in the 30-mg group was attributed to severe cerebral infarction and was considered unlikely to be related to HRS-7535 or metformin.

[b] Includes liver function abnormalities, amylase and/or lipase levels greater than 3 times the ULN, acute and/or chronic pancreatitis, abnormal blood calcitonin level, and episodes of severe hypoglycemia.

[c] Elevated aspartate aminotransferase (AST) and/or alanine aminotransferase (ALT) level and total bilirubin (TBIL) level were reported as AESIs if the following conditions were met: (1) AST and ALT levels were normal at baseline and exceeded 3 times ULN during the 16-week treatment period and, if above normal at baseline, levels had to exceed 2 times baseline and more than 3 times ULN or more than 8 times ULN (whichever was lower); (2) TBIL level was normal at baseline and exceeded 2 times ULN and, if above normal at baseline, the increase had to be greater than 1 times ULN or exceed 3 times ULN (whichever was lower); and (3) no hemolysis and alkaline phosphatase levels less than 2 times ULN.

[d] Defined as a postbaseline calcitonin concentration 50 ng/L or greater with an increase greater than 50% from baseline.

[e] Defined as those occurring after study drug administration and meeting level 3 hypoglycemia criteria, according to the American Diabetes Association 2020 guidelines.

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was low; all events were level 1, and no level 2 or level 3 hypoglycemia was observed, consistent with the glucose-dependent mechanism of GLP-1 RAs.[26,31,32]

Mild to moderate increases in pancreatic enzyme levels and small increases in calcitonin levels were observed with HRS-7535, but no cases of pancreatitis, calcitonin-related AEs of special interest, or alanine aminotransferase or aspartate aminotransferase elevations greater than 3 times the upper limit of normal were identified. These findings are consistent with data from large GLP-1 RA development programs, in which transient elevations of pancreatic enzyme levels are common and not considered to be associated with pancreatitis, whereas medullary thyroid carcinoma risk has remained low.[11,12,18,22,29-31,33-35] Given prior concerns about elevated liver enzyme levels with lotiglipron,[19] hepatic monitoring was prioritized in this trial; however, no hepatic AEs of special interest were observed. Increases in pulse and heart rate with HRS-7535 were modest and consistent with known class effects.[26,33]

Limitations

This study has several limitations. First, the double-blind treatment period was limited to 16 weeks, with exposure to the target 60-mg or 90-mg dose for only 8 to 12 weeks. Longer-term studies are needed to characterize durability of glycemic control and weight reduction and the long-term safety of HRS-7535. Second, the trial enrolled adults in China with relatively low baseline body weight and BMI compared with populations in many global type 2 diabetes trials, which may limit generalizability.

The OUTSTAND (Oral Small Molecule GLP-RA HRS-7535 Trial Series Targeting Metabolism in Diabetes) phase 3 program includes two 52-week trials[36,37] evaluating HRS-7535 in patients with inadequate glycemic control despite lifestyle intervention or other glucose-lowering therapies. These studies may further inform dose selection and the longer-term benefit-risk profile of HRS-7535 in broader populations with type 2 diabetes.

Conclusions

In this randomized clinical trial of adults in China with type 2 diabetes inadequately controlled by metformin monotherapy, 16 weeks of once-daily oral HRS-7535 added to metformin improved HbA_{1c} levels and other glycemic measures and was associated with modest weight reduction. HRS-7535 was generally well tolerated, with a safety profile consistent with that of GLP-1 RAs. As a nonpeptide oral GLP-1 RA that does not require fasting administration or injection, HRS-7535 may provide a convenient treatment option, pending confirmation in phase 3 trials.

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Author Contributions: Prof Guo had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

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Supervision: Guo.

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SUPPLEMENT 1.

Trial Protocol and Statistical Analysis Plan

SUPPLEMENT 2.

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SUPPLEMENT 3.

Data Sharing Statement

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